



Examination Number:

Set:

# Shore

## Year 12

### HSC Assessment Task 3

### Term II Examination

### 2016

# Mathematics Extension 1

#### General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- The BOSTES Reference Sheet is provided separately
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–14 show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–14 in a new writing booklet
- Write your examination number on the front cover of each booklet

If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total marks – 70

#### Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### Section II Pages 6–11

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 The point  $P(x, y)$  divides the interval joining  $A(-3, 4)$  to  $B(7, 9)$  internally in the ratio  $2:3$ .

What are the coordinates of  $P$ ?

(A)  $(1, 6)$

(B)  $(1, 5)$

(C)  $(0, 6)$

(D)  $(0, 5)$

- 2 For what value of  $b$  is  $(x+1)$  a factor of  $x^3 - bx + 3$ ?

(A)  $b = 3$

(B)  $b = -3$

(C)  $b = 2$

(D)  $b = -2$

- 3 What is the acute angle between the lines  $x + y = 1$  and  $2x - y = 3$ ?

(A)  $18^\circ$

(B)  $19^\circ$

(C)  $71^\circ$

(D)  $72^\circ$

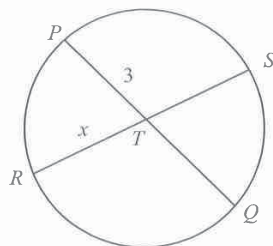
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- 4 The point  $P(x, y)$  moves so that it is equidistant from the  $y$ -axis and the point  $S(2, 0)$ .

What is the equation of the locus of  $P$ ?

- (A)  $y^2 = 4(x - 2)$   
 (B)  $y^2 = 4(x - 1)$   
 (C)  $y^2 = 8(x - 2)$   
 (D)  $y^2 = 8(x - 1)$

5



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SCALE

In the circle, chords  $PQ$  and  $RS$  intersect at  $T$ ,  $PQ = 7$ ,  $PT = 3$ ,  $RS = 8$  and  $RT = x$ .

What are the possible values of  $x$ ?

- (A)  $x = 6$  or  $x = 2$   
 (B)  $x = 4$  or  $x = 2$   
 (C)  $x = 6$  or  $x = 4$   
 (D)  $x = 4$  or  $x = 3$
- 6 What is the domain for  $f(x) = \sqrt{1-x} + \sqrt{1+x}$ ?
- (A)  $x > 1$  or  $x < -1$   
 (B)  $x \geq 1$  or  $x \leq -1$   
 (C)  $-1 < x < 1$   
 (D)  $-1 \leq x \leq 1$

- 7 A curve has parametric equations  $x = 3 \cos t$  and  $y = 2 \sin t$ .

What is the Cartesian equation of the curve?

- (A)  $2x^2 + 3y^2 = 6$   
 (B)  $2x^2 - 3y^2 = 6$   
 (C)  $4x^2 + 9y^2 = 36$   
 (D)  $4x^2 - 9y^2 = 36$

- 8 The angle  $\theta$  satisfies  $\sin \theta = \frac{3}{5}$  and  $\frac{\pi}{2} \leq \theta \leq \pi$ .

What is the value of  $\sin 2\theta$ ?

- (A)  $\frac{24}{25}$   
 (B)  $-\frac{24}{25}$   
 (C)  $\frac{12}{25}$   
 (D)  $-\frac{12}{25}$

- 9 Which one of the following is the primitive (indefinite integral) of  $\frac{1}{4x^2 + 16}$ ?

- (A)  $\frac{1}{8} \tan^{-1} \frac{x}{2} + C$   
 (B)  $\frac{1}{4} \tan^{-1} \frac{x}{2} + C$   
 (C)  $\frac{1}{2} \tan^{-1} 2x + C$   
 (D)  $\frac{1}{4} \tan^{-1} 2x + C$

## Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

**Question 11 (15 marks)** Use a SEPARATE writing booklet

(a) Evaluate  $\lim_{x \rightarrow 0} \frac{\sin x}{3x}$ . 1

(b) Solve  $\frac{2x}{1+x} \leq 1$ . 2

(c) Find  $\int \cos^2 x \, dx$ . 2

(d) By using the substitution  $\tan \frac{A}{2} = t$ , or otherwise, show that 2

$$\frac{1 + \sin A + \cos A}{1 + \sin A - \cos A} = \cot \frac{A}{2}.$$

(e) Find  $\frac{d}{dx} [\cos^{-1}(x^2)]$ . 2

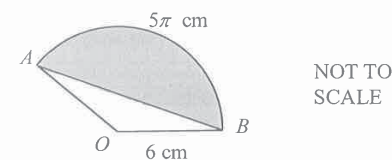
(f) (i) Write down the domain and range of the function  $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$ . 2

(ii) Sketch the graph of  $y = f(x)$ . 1

(g) Use the substitution  $u = x^2 - 1$  to evaluate  $\int_2^3 \frac{x}{(x^2 - 1)^2} \, dx$ . 3

**Question 12 (15 marks)** Use a SEPARATE writing booklet

(a)



3

$AOB$  is a sector of a circle with centre  $O$ .  
The length of the arc  $AB$  is  $5\pi$  centimetres.

Find the exact area of the segment cut off by the arc  $AB$  and the chord  $AB$ .

(b) Simplify  $\frac{4 \sin 2\theta \cos 2\theta}{\cos 4\theta}$ . 2

(c) (i) Express  $\sin x + \sqrt{3} \cos x$  in the form  $R \sin(x + \theta)$  2  
where  $R > 0$  and  $0 < \theta < \frac{\pi}{2}$ .

(ii) Hence or otherwise solve the equation  $\sin x + \sqrt{3} \cos x = \sqrt{3}$  for  $0 \leq x \leq 2\pi$ . 2

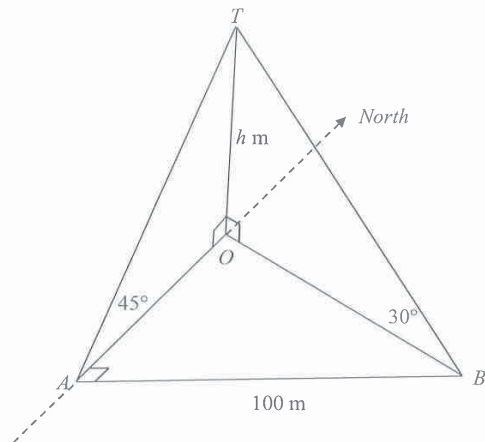
(d) Find the general solutions to  $\cos\left(2x - \frac{\pi}{3}\right) = 0$ . 3

(e) Prove by mathematical induction that for all integer values of  $n \geq 1$ , 3

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

**Question 13 (15 marks)** Use a SEPARATE writing booklet

(a)



A surveyor stands at a point  $A$  which is due south of a tower  $OT$  of height  $h$  metres. The angle of elevation of the top of the tower from  $A$  is  $45^\circ$ . The surveyor then walks due east for 100 metres to a point  $B$ , where he observes the top of the tower to have an angle of elevation of  $30^\circ$ .

- |                                                                                                              |   |
|--------------------------------------------------------------------------------------------------------------|---|
| (i) Show that $OB = h\sqrt{3}$ .                                                                             | 1 |
| (ii) Show that $h = 50\sqrt{2}$ .                                                                            | 2 |
| (iii) Find the bearing of $B$ from the base of the tower.<br>Give your answer correct to the nearest degree. | 2 |

(b) The polynomial  $P(x) = x^3 + px^2 + qx + r$  has roots  $-\sqrt{\alpha}$ ,  $\sqrt{\alpha}$  and  $\beta$ .

- |                                    |   |
|------------------------------------|---|
| (i) Show that $\beta + p = 0$ .    | 1 |
| (ii) Show that $\alpha\beta = r$ . | 1 |
| (iii) Show that $pq = r$ .         | 2 |

**Question 13 continues on page 9**

**Question 13 (continued)**

(c) The points  $P(8p, 4p^2)$  and  $Q(8q, 4q^2)$  lie on the parabola  $x^2 = 16y$ .

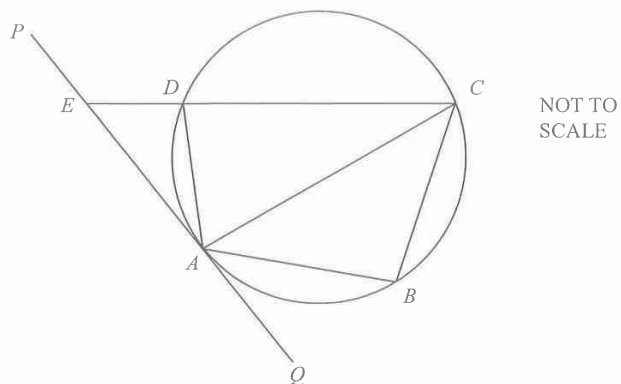
- |                                                                                                     |   |
|-----------------------------------------------------------------------------------------------------|---|
| (i) Derive the equation of the chord $PQ$ . Write your answer in general form.                      | 3 |
| (ii) Show that if $PQ$ is a focal chord then $pq = -1$ .                                            | 1 |
| (iii) If $PQ$ is a focal chord and $P$ has coordinates $(4, 1)$ , what are the coordinates of $Q$ ? | 2 |

**End of Question 13**

**Question 14 (15 marks)** Use a SEPARATE writing booklet

- (a) In the diagram,  $PQ$  is a tangent to the circle at  $A$ .  
Points  $B$ ,  $C$  and  $D$  lie on the circle.

2



Copy or trace the diagram into your writing booklet.

Show, giving reasons, that  $\angle EAC = \angle ADE$ .

- (b) (i) Find the range of the function  $y = \frac{1}{\sqrt{4-x^2}}$ , 1
- (ii) Find the exact area between the curve  $y = \frac{1}{\sqrt{4-x^2}}$ , the  $x$  axis and the lines  $x = -\sqrt{3}$  and  $x = \sqrt{2}$ . 2
- (c) Let  $f(x) = \sin^{-1}(-x) + \cos^{-1}(-x)$ , where  $-1 \leq x \leq 1$ .
- (i) Show that  $f'(x) = 0$ . 2
- (ii) Hence deduce that  $\sin^{-1}(-x) + \cos^{-1}(-x) = \frac{\pi}{2}$ . 2

**Question 14 continues on page 11**

**Question 14 (continued)**

- (d) Given  $f(x) = \log_e \left( \frac{2-x}{x} \right)$  for  $0 < x < 2$ .

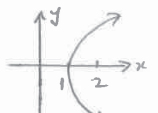
- (i) Find  $f'(x)$ . Express your answer as a single fraction. 2
- (ii) Explain why  $f(x)$  has an inverse function for  $0 < x < 2$ . 2
- (iii) Find the equation of the inverse function,  $f^{-1}(x)$ . 2

**END OF PAPER**

$$\begin{aligned} \textcircled{1} \quad P(x, y) &= \left( \frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right) \\ &= \left( \frac{2(7) + 3(-3)}{2+3}, \frac{2(9) + 3(4)}{2+3} \right) \\ &= \left( \frac{5}{5}, \frac{30}{5} \right) \\ &= \underline{(1, 6)} \quad \textcircled{A} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad P(-1) &= 0 = (-1)^3 - b(-1) + 3 \\ 0 &= -1 + b + 3 \\ \underline{b} &= \underline{-2} \quad \textcircled{D} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \tan \alpha &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-1 - 2}{1 + (-1)(2)} \right| \\ &= \left| \frac{-3}{-1} \right| \\ &= 3 \\ \alpha &= \tan^{-1} 3 \\ &= \underline{72^\circ} \quad \textcircled{D} \end{aligned}$$

$$\textcircled{4} \quad \begin{aligned} &\text{Graph of } y^2 = 4(x-1) \end{aligned}$$


$\underline{y^2 = 4(x-1)}$   $\textcircled{B}$

$$\begin{aligned} \textcircled{5} \quad \text{PT. } TQ &= \text{RT. } RS \\ 3 \times 4 &= x(8-x) \\ 12 &= 8x - x^2 \\ x^2 - 8x + 12 &= 0 \\ (x-6)(x-2) &= 0 \\ \underline{x=6} \text{ or } \underline{x=2} \quad \textcircled{A} \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad f(x) &= \sqrt{1-x} + \sqrt{1+x} \\ 1-x &\geq 0 \text{ and } 1+x \geq 0 \\ x &\leq 1 \text{ and } x \geq -1 \\ \text{i.e. } \underline{-1 \leq x \leq 1} \quad \textcircled{D} \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad x &= 3 \cos t \quad y = 2 \sin t \\ \frac{x}{3} &= \cos t \quad \frac{y}{2} = \sin t \\ \cos^2 t + \sin^2 t &= 1 \\ \left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 &= 1 \\ \frac{x^2}{9} + \frac{y^2}{4} &= 1 \\ \underline{4x^2 + 9y^2} &= \underline{36} \quad \textcircled{C} \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad \sin \theta &= \frac{3}{5} \quad \cos \theta = -\frac{4}{5} \\ \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 \times \frac{3}{5} \times -\frac{4}{5} \\ &= \underline{-\frac{24}{25}} \quad \textcircled{B} \end{aligned}$$

$$\begin{aligned} \textcircled{9} \quad \int \frac{1}{4x^2 + 16} dx &= \frac{1}{4} \int \frac{1}{x^2 + 4} dx \\ &= \frac{1}{4} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + C \\ &= \underline{\frac{1}{8} \tan^{-1} \frac{x}{2} + C} \quad \textcircled{A} \end{aligned}$$

$$\begin{aligned} \textcircled{10} \quad \text{Odd function in 2nd quadrant} \\ \therefore y &= \underline{-\frac{\pi}{1+x^2}} \quad \textcircled{D} \end{aligned}$$

A D D B A D C B A D

### Question 11

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow 0} \frac{\sin x}{3x} &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{1}{3} \\ &= 1 \times \frac{1}{3} \\ &= \underline{\underline{\frac{1}{3}}} \quad [1] \end{aligned}$$

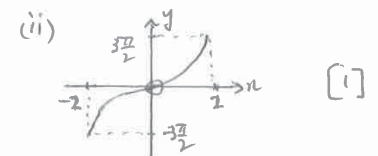
$$\begin{aligned} \text{(b)} \quad \frac{2x}{1+x} &\leq 1 \\ \text{Critical Points: } x &\neq -1, \quad \frac{2x}{1+x} = 1 \\ 2x &= 1+x \\ x &= 1 \\ \text{Graph: } x &\text{ from } -1 \text{ to } 1 \text{ with a checkmark at } x=1. \\ \therefore \underline{-1 < x \leq 1} \quad [2] \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \int \cos^2 x \, dx &= \int \frac{1}{2} (\cos 2x + 1) \, dx \\ &= \frac{1}{2} \left( \frac{1}{2} \sin 2x + x \right) + C \\ &= \underline{\underline{\frac{1}{4} \sin 2x + \frac{x}{2} + C}} \quad [2] \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad \text{LHS} &= \frac{1 + \sin A + \cos A}{1 + \sin A - \cos A} \\ &= \frac{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \\ &= \frac{1+t^2+2t+1-t^2}{1+t^2+2t-(1-t^2)} \\ &= \frac{2t+2}{2t^2+2t} \\ &= \frac{2(t+1)}{2t(t+1)} \\ &= \frac{1}{t} \\ &= \cot \frac{A}{2} = \underline{\underline{\text{RHS}}} \quad [2] \end{aligned}$$

$$\begin{aligned} \text{(e)} \quad \frac{d}{dx} [\cos^{-1}(x^2)] &= \frac{-1}{\sqrt{1-(x^2)^2}} \times 2x \\ &= \underline{\underline{\frac{-2x}{\sqrt{1-x^4}}}} \quad [2] \end{aligned}$$

$$\begin{aligned} \text{(f)} \quad f(x) &= 3 \sin^{-1}\left(\frac{x}{2}\right) \\ \text{(i)} \quad D: \quad &-1 \leq \frac{x}{2} \leq 1 \\ &\underline{-2 \leq x \leq 2} \quad [2] \\ R: \quad &-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{\pi}{2} \\ &\underline{\underline{-\frac{3\pi}{2} \leq 3 \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{3\pi}{2}}} \end{aligned}$$



$$\begin{aligned} \text{(g)} \quad \int_2^3 \frac{x}{(x^2-1)^2} dx &\quad u = x^2 - 1 \\ &\quad du = 2x \, dx \\ &\quad \frac{1}{2} du = x \, dx \\ &= \int_3^8 \frac{du}{2u^2} \quad \begin{array}{l} x=3 \rightarrow u=8 \\ x=2 \rightarrow u=3 \end{array} \\ &= \frac{1}{2} \left[ \frac{u^{-1}}{-1} \right]_3^8 \\ &= -\frac{1}{2} \left[ \frac{1}{u} \right]_3^8 \\ &= -\frac{1}{2} \left[ \frac{1}{8} - \frac{1}{3} \right] \\ &= -\frac{1}{2} \times -\frac{5}{24} \\ &= \underline{\underline{\frac{5}{48}}} \end{aligned}$$

### Question 12

(a)  $l = r\theta$

$5\pi = 6\theta$

$\theta = \frac{5\pi}{6}$

$$\begin{aligned} \text{Area} &= \frac{1}{2}r^2(\theta - \sin\theta) \\ &= \frac{1}{2} \times 6^2 \left( \frac{5\pi}{6} - \sin\frac{5\pi}{6} \right) \\ &= 18 \left( \frac{5\pi}{6} - \frac{1}{2} \right) \\ &= \underline{\underline{(15\pi - 9) \text{ cm}^2}} \quad [3] \end{aligned}$$

(b)  $\frac{4 \sin 2\theta \cos 2\theta}{\cos 4\theta}$

$$\begin{aligned} &= \frac{2 \times 2 \sin 2\theta \cos 2\theta}{\cos 4\theta} \\ &= \frac{2 \sin 4\theta}{\cos 4\theta} \quad [2] \\ &= \underline{\underline{2 \tan 4\theta}} \end{aligned}$$

(c)

(i)  $\sin x + \sqrt{3} \cos x = R \sin(x + \theta)$

$$= R \sin x \cos \theta + R \cos x \sin \theta$$

$\therefore R \cos \theta = 1 \quad R \sin \theta = \sqrt{3}$

$$\cos \theta = \frac{1}{R} \quad \sin \theta = \frac{\sqrt{3}}{R}$$

$\sqrt{3} \quad R \quad \theta \quad 1$

$$\begin{aligned} R^2 &= 1^2 + (\sqrt{3})^2 \quad \cos \theta = \frac{1}{2} \\ &= 1 + 3 \\ &= 4 \\ \underline{R} &= \underline{2} \end{aligned}$$

$\therefore \underline{\underline{\sin x + \sqrt{3} \cos x = 2 \sin(x + \frac{\pi}{3})}} \quad [2]$

(ii)  $2 \sin(x + \frac{\pi}{3}) = \sqrt{3}$

$\sin(x + \frac{\pi}{3}) = \frac{\sqrt{3}}{2}$

$x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \dots$

for  $0 \leq x \leq 2\pi \quad x = 0, \frac{\pi}{3}, 2\pi \quad [2]$

(d)  $\cos(2x - \frac{\pi}{3}) = 0$

$2x - \frac{\pi}{3} = 2n\pi \pm \cos^{-1}0$   
(where  $n$  is an integer)

$2x - \frac{\pi}{3} = 2n\pi + \frac{\pi}{2} \quad 2x - \frac{\pi}{3} = 2n\pi - \frac{\pi}{2}$

$2x = 2n\pi + \frac{5\pi}{6} \quad 2x = 2n\pi - \frac{\pi}{6}$

$x = n\pi + \frac{5\pi}{12} \quad x = n\pi - \frac{\pi}{12} \quad [3]$

(e)  $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

Prove true for  $n=1$

$\text{LHS} = \frac{1}{1 \times 3} \quad \text{RHS} = \frac{1}{2 \times 1 + 1}$

$$= \frac{1}{3} \quad = \frac{1}{3}$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

i.e.  $S_k = \frac{k}{2k+1}$

Prove true for  $n=k+1$

i.e.  $S_k + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = \frac{k+1}{2(k+1)+1}$

i.e.  $S_k + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$

Q12 (e) continued

$\text{LHS} = \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$

$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$

$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$

$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$

$= \frac{k+1}{2k+3}$

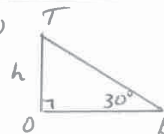
$= \text{RHS}$

$\therefore$  If true for  $n=k$ , then true for  $n=k+1$ .

By process of induction, true for all integers,  $n \geq 1$ . [3]

### Question 13

(a) (i)



$\frac{OB}{h} = \tan 60^\circ$

$OB = h \tan 60^\circ$

$= \underline{\underline{h\sqrt{3}}} \quad [1]$

(ii) In  $\triangle OAT$ ,  $OA = h$

In  $\triangle OAB$ ,  $OA^2 + 100^2 = OB^2$

$h^2 + 10000 = (h\sqrt{3})^2$

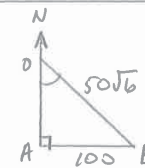
$h^2 + 10000 = 3h^2$

$10000 = 2h^2$

$h^2 = 5000$

$h = \underline{\underline{50\sqrt{2}}} \quad [2]$

(iii)



$OB = h\sqrt{3}$

$= 50\sqrt{2} \times \sqrt{3}$

$= \underline{\underline{50\sqrt{6}}}$

$\sin \angle AOB = \frac{100}{50\sqrt{6}}$

$= \frac{2}{\sqrt{6}}$

$\angle AOB = \underline{\underline{55^\circ}} \quad [2]$

$\therefore$  Bearing is  $180^\circ - 55^\circ = \underline{\underline{125^\circ}}$

(b) (i)  $-\sqrt{x} + \sqrt{x} + \beta = \frac{-p}{1}$

$\beta = -p$

$\underline{\underline{\beta + p = 0}} \quad [1]$

(ii)  $-\sqrt{x} \cdot \sqrt{x} \cdot \beta = \frac{-r}{1}$

$-x\beta = -r$

$\underline{\underline{\alpha\beta = r}} \quad [1]$

(iii)  $-\sqrt{x} \cdot \sqrt{x} + \sqrt{x} \cdot \beta + \sqrt{x} \cdot \beta = q$

$-x = q$

$\underline{\underline{\alpha = -q}}$

sub into (ii)

$-q\beta = r$

$\underline{\underline{\beta = \frac{r}{-q}}}$

sub into (i)

$\frac{r}{-q} + p = 0$

$r - pq = 0$

$\underline{\underline{pq = r}} \quad [2]$

$$(c) i) m = \frac{4p^2 - 4q^2}{8p - 8q}$$

$$= \frac{4(p-q)(p+q)}{8(p-q)}$$

$$= \frac{p+q}{2}$$

Eg'n of PQ is:

$$y - 4q^2 = \frac{p+q}{2}(x - 8q)$$

$$y - 4q^2 = \left(\frac{p+q}{2}\right)x - 4pq - 4q^2$$

$$\left(\frac{p+q}{2}\right)x - y - 4pq = 0 \quad [3]$$

(ii) PQ passes through (0, 4)

$$\therefore \left(\frac{p+q}{2}\right) \times 0 - 4 - 4pq = 0$$

$$-4pq = 4$$

$$\underline{pq = -1} \quad [1]$$

(iii)  $P(8p, 4p^2) \equiv (4, 1)$

$$\therefore 8p = 4$$

$$p = \frac{1}{2} \rightarrow q = -2 \quad (pq = -1)$$

$\therefore Q$  is  $(8x-2, 4(-2)^2)$

$$= \underline{(-16, 16)} \quad [2]$$

### Question 14

$$(a) \angle EAC = \angle ABC$$

( $\angle$  between chord + tangent equals  $\angle$  in alternate segment)

$$\angle ABC = \angle ADE$$

( $\angle$  in cyclic quadrilateral equals opposite exterior angle)

$$\therefore \underline{\angle EAC = \angle ADE} \quad (\text{both} = \angle ABC) \quad [2]$$

$$(b) (i) \text{ Range: } 4 - x^2 \leq 4$$

$$0 \leq \sqrt{4 - x^2} \leq 2$$

$$\underline{\underline{\frac{1}{\sqrt{4 - x^2}} \geq \frac{1}{2}}} \quad [1]$$

$$(ii) A = \int_{-\sqrt{3}}^{\sqrt{2}} \frac{1}{\sqrt{4 - x^2}} dx$$

$$= \left[ \sin^{-1} \frac{x}{2} \right]_{-\sqrt{3}}^{\sqrt{2}}$$

$$= \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{4} - -\frac{\pi}{3}$$

$$= \underline{\underline{\frac{7\pi}{12} \text{ units}^2}} \quad [2]$$

$$(c) (i) f(x) = \sin^{-1}(-x) + \cos^{-1}(-x)$$

$$f'(x) = \frac{1}{\sqrt{1 - (-x)^2}} \times -1 + \frac{-1}{\sqrt{1 - (-x)^2}} \times -1$$

$$= \frac{-1}{\sqrt{1 - x^2}} + \frac{1}{\sqrt{1 - x^2}}$$

$$\underline{\underline{= 0}} \quad [2]$$

$$(c) (ii) \text{ If } f'(x) = 0$$

then  $f(x) = \text{constant}$

$$\text{let } x=0 \quad f(0) = \sin^{-1}(0) + \cos^{-1}(0)$$

$$= 0 + \frac{\pi}{2}$$

$$= \underline{\underline{\frac{\pi}{2}}}$$

$$\therefore \underline{\underline{\sin^{-1}(-x) + \cos^{-1}(-x) = \frac{\pi}{2}}}$$

[2]

$$(d) (i) f(x) = \ln\left(\frac{2-x}{x}\right)$$

$$= \ln(2-x) - \ln x$$

$$f'(x) = \frac{-1}{2-x} - \frac{1}{x}$$

$$= \frac{-x - (2-x)}{(2-x)x}$$

$$= \underline{\underline{\frac{-2}{(2-x)x}}} \quad [2]$$

$$(ii) \text{ For } 0 < x < 2, (2-x)x > 0$$

$$\therefore \frac{-2}{(2-x)x} < 0$$

$$\text{i.e. } f'(x) < 0$$

i.e.  $f(x)$  is monotonic decreasing

$\therefore f(x)$  has an inverse for  $0 < x < 2$

[2]

$$(iii) \text{ let } y = \ln\left(\frac{2-x}{x}\right)$$

$$\text{Inverse} \Rightarrow x = \ln\left(\frac{2-y}{y}\right)$$

$$e^x = \frac{2-y}{y}$$

$$ye^x = 2-y$$

$$ye^x + y = 2$$

$$y(e^x + 1) = 2$$

$$y = \frac{2}{e^x + 1}$$

$$\therefore \underline{\underline{f^{-1}(x) = \frac{2}{e^x + 1}}}$$

[2]